Reflection and Refraction

Introduction to the laws of reflection and refraction in linear dielectrics

Vivek Kalita

Department of Physics, IISER Mohali

10 November 2026

Contents

Maxwell's equations and the boundary conditions 1

1.1 Boundary conditions at a medium-discontinuity 2

Plane Waves 3

2.1 Laws of reflection andd refraction 4

1 Maxwell's equations and the boundary conditions

Let's start with the Maxwell's equations:

Axiom 1.1 (Maxwell's equations).

The four fundamental equations for the electromagnetic field are

$$\iint_{\partial V} \mathbf{D}\cdot\mathrm{d}\mathbf{A} = \iiint_{V} \rho_f \,\mathrm{d}V \qquad \text{(A1)} \qquad \oint_{\partial S} \mathbf{E}\cdot\mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_{S} \mathbf{B}\cdot\mathrm{d}\mathbf{A} \qquad \text{(A2)}$$

$$\iint_{\partial V} \mathbf{B}\cdot\mathrm{d}\mathbf{A} = 0 \qquad \text{(A3)} \qquad \oint_{\partial S} \mathbf{H}\cdot\mathrm{d}\mathbf{l} = \iint_{S} \mathbf{J}_f\cdot\mathrm{d}\mathbf{A} + \frac{\mathrm{d}}{\mathrm{d}t}\iint_{S} \mathbf{D}\cdot\mathrm{d}\mathbf{A} \qquad \text{(A4)}$$

The constitutive relations are

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} \qquad \mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}.$$

1.1 Boundary conditions at a medium-discontinuity

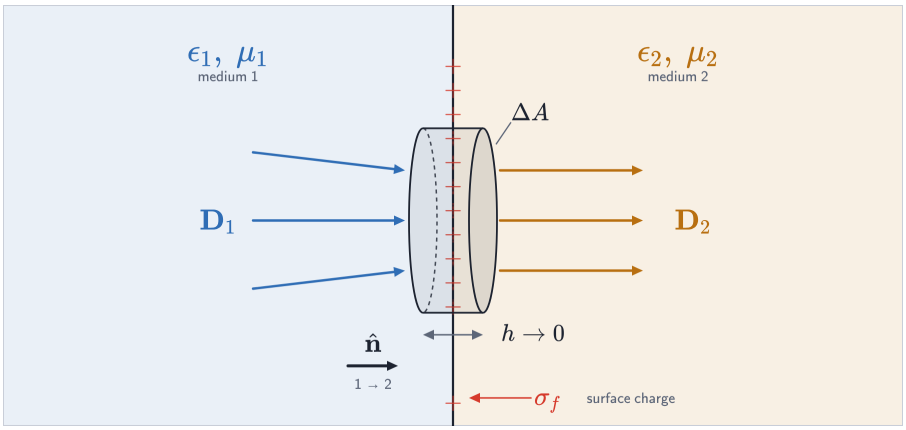


Figure 1: The Gaussian pillpox

1. As illustrated in the figure, consider a Gaussian pillbox of cross-sectional area A and height h at the interface between media 1 and 2. Let $\mathbf{n}$ be the unit normal directed from medium 1 to medium 2. By Gauss's law for the electric displacement field,

$$\oint_{\partial V} \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}} = \sigma_f A,$$

thus,

$$\mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f.$$

2. Applying Gauss's law for the magnetic flux density to the same pillbox gives

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{a} = 0 \quad \Longrightarrow \quad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.$$

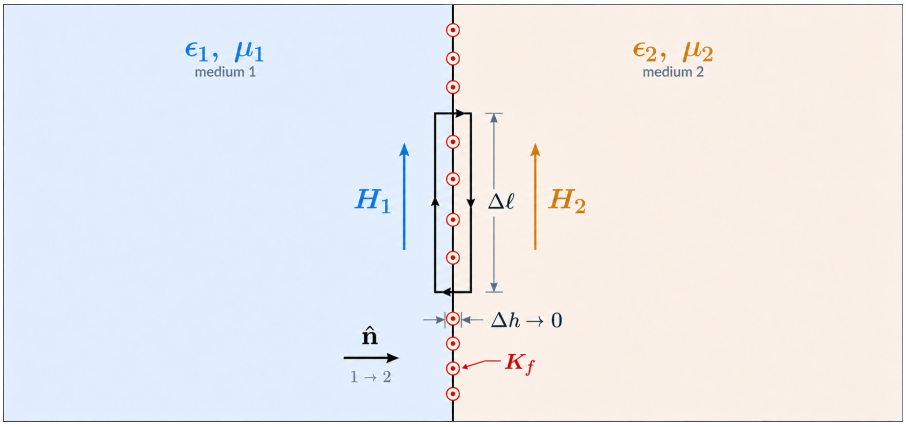


Figure 2: The amperian loops

1. Consider an infinitesimal rectangular loop on the interface. We have:

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

As the loop height tends to zero, the magnetic flux through its area vanishes. So the tangential component of the electric field is continuous:

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}.$$

2. Applying the Ampère–Maxwell law to the same loop gives

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f,\text{enc}} + \frac{d}{dt} \int_S \mathbf{D} \cdot d\mathbf{a}.$$

As the loop height tends to zero, the displacement-flux term vanishes:

$$\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_{s,f}.$$

Theorem 1.2 (Boundary conditions for discontinuous media).

Let $\mathbf{n}$ *be the unit normal directed from medium 1 to medium 2. The boundary conditions are*

$$\begin{aligned} \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) &= \mathbf{0}, & \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) &= \mathbf{K}_f, \\ \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= \sigma_f, & \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) &= 0. \end{aligned}$$

Corollary 1.3 (Boundary conditions for linear isotropic media).

For linear, isotropic media,

$$\mathbf{D} = \varepsilon \mathbf{E}, \qquad \mathbf{B} = \mu \mathbf{H}.$$

Therefore, at the interface,

$$\begin{aligned} \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) &= \mathbf{0}, & \mathbf{n} \times \left(\frac{\mathbf{B}_2}{\mu_2} - \frac{\mathbf{B}_1}{\mu_1} \right) &= \mathbf{K}_f, \\ \mathbf{n} \cdot (\varepsilon_2 \mathbf{E}_2 - \varepsilon_1 \mathbf{E}_1) &= \sigma_f, & \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) &= 0. \end{aligned}$$

If there is no free surface current or surface charge density, the boundary conditions reduce to

$$\begin{aligned} E_{2\parallel} &= E_{1\parallel}, & \frac{B_{2\parallel}}{\mu_2} &= \frac{B_{1\parallel}}{\mu_1}, \\ \varepsilon_2 E_{2\perp} &= \varepsilon_1 E_{1\perp}, & B_{2\perp} &= B_{1\perp}. \end{aligned}$$

2 Plane Waves

Consider a plane wave of the form

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}.$$

From Faraday's law we could relate the electric and the magnetic field as:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \Longrightarrow \quad \mathbf{B} = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}.$$

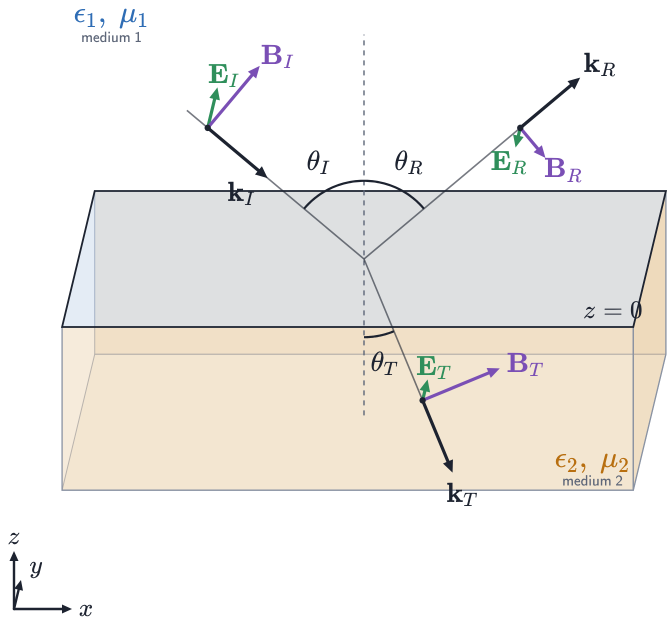


Figure 3: A plane wave incident on a surface (the xy plane)

Let the wave vectors for the incident, reflected, and transmitted waves be

$$\mathbf{k}_i = k_1 \left(\sin\theta_i\, \hat{\mathbf{x}} - \cos\theta_i\, \hat{\mathbf{z}}\right) \quad \mathbf{k}_r = k_1 \left(\sin\theta_r\, \hat{\mathbf{x}} + \cos\theta_r\, \hat{\mathbf{z}}\right),$$

$$\mathbf{k}_t = k_2 \left(\sin\theta_t\, \hat{\mathbf{x}} - \cos\theta_t\, \hat{\mathbf{z}}\right).$$

Assume the electric fields are polarized along the y-axis:

$$\mathbf{E}_i(\mathbf{r},t) = E_0 \hat{\mathbf{y}}\, e^{i(\mathbf{k}_i\cdot\mathbf{r}-\omega t)} \quad \mathbf{E}_r(\mathbf{r},t) = E_{r0} \hat{\mathbf{y}}\, e^{i(\mathbf{k}_r\cdot\mathbf{r}-\omega t)},$$

$$\mathbf{E}_t(\mathbf{r},t) = E_{t0} \hat{\mathbf{y}}\, e^{i(\mathbf{k}_t\cdot\mathbf{r}-\omega t)}.$$

Using the relation between the magnetic field and the electric field:

$$\mathbf{B}_i(\mathbf{r},t) = \frac{k_1 E_0}{\omega} \left(\cos\theta_i\, \hat{\mathbf{x}} + \sin\theta_i\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_i\cdot\mathbf{r}-\omega t)} \quad \mathbf{B}_r(\mathbf{r},t) = \frac{k_1 E_{r0}}{\omega} \left(-\cos\theta_r\, \hat{\mathbf{x}} + \sin\theta_r\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_r\cdot\mathbf{r}-\omega t)}$$

$$\mathbf{B}_t(\mathbf{r},t) = \frac{k_2 E_{t0}}{\omega} \left(\cos\theta_t\, \hat{\mathbf{x}} + \sin\theta_t\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_t\cdot\mathbf{r}-\omega t)}.$$

2.1 Laws of reflection andd refraction

At the interface $z = 0$, continuity of the tangential electric field gives

$$E_{i\parallel} + E_{r\parallel} = E_{t\parallel}.$$

Since all three electric fields are polarized along the y-axis, this becomes

$$E_0 e^{i(k_1 \sin\theta_i\, x-\omega t)} + E_{r0} e^{i(k_1 \sin\theta_r\, x-\omega t)} = E_{t0} e^{i(k_2 \sin\theta_t\, x-\omega t)}.$$

1. Since the boundary condition must hold for every x, and all three waves have the same frequency, their tangential phase components must match:

   $$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t.$$

   The first equality gives

   $$\theta_i = \theta_r,$$

   which is the law of reflection.

2. The second equality gives

   $$k_1 \sin\theta_i = k_2 \sin\theta_t.$$

   For a plane wave in a homogeneous isotropic medium,

   $$k_j = \frac{n_j \omega}{c}.$$

   where n_j is the refractive index. Since the frequency is the same in both media, this becomes

   $$n_1 \sin\theta_i = n_2 \sin\theta_t,$$

   which is Snell's law.

Theorem 2.1 (Law of reflection and Snell's law).

- *According to the* law of reflection*, if θ_i and θ_r are the angles of incidence and reflection, respectively, then*

  $$\theta_i = \theta_r.$$

- *For an interface between two media with refractive indices n_1 and n_2,* Snell's law *is*

  $$n_1 \sin\theta_i = n_2 \sin\theta_t,$$

  where θ_i and θ_t are the angles of incidence and refraction, respectively.